

Experiment 2: Binary Search (Recursive)

Aim

To implement the **Binary Search (Recursive)** algorithm using Python to search for an element in a sorted array.

Algorithm

Step 1: Start.

Step 2: Read the number of elements n .

Step 3: Read the sorted array elements.

Step 4: Read the key element to be searched.

Step 5: Set $low = 0$ and $high = n - 1$.

Step 6: Find the middle element:

- $mid = (low + high) // 2$

Step 7:

- If $arr[mid] == key$, display the position.
- If $key < arr[mid]$, search the left half recursively.
- If $key > arr[mid]$, search the right half recursively.

Step 8: If $low > high$, display "**Element not found**".

Step 9: Stop.

Source Code (Python)

main.py    Share 

```
1 # Binary Search using Recursion
2
3 def binary_search(arr, low, high, key):
4     if low <= high:
5         mid = (low + high) // 2
6
7         if arr[mid] == key:
8             return mid
9         elif key < arr[mid]:
10            return binary_search(arr, low, mid - 1, key)
11        else:
12            return binary_search(arr, mid + 1, high, key)
13
14    return -1
15
16
17 n = int(input("Enter number of elements: "))
18
19 arr = []
20
21 print("Enter sorted elements:")
22 for i in range(n):
23     arr.append(int(input()))
24
25 key = int(input("Enter element to search: "))
26
27 result = binary_search(arr, 0, n - 1, key)
28
29 if result != -1:
30     print("Element found at index", result)
31 else:
32     print("Element not found")
```

Output

Output

```
Enter number of elements: 6
Enter sorted elements:
10
20
30
40
50
60
Enter element to search: 40
Element found at index 3

=== Code Execution Successful ===
```

Time Complexity

- **Best Case (Ω): $O(1)$**
- **Average Case (Θ): $O(\log n)$**
- **Worst Case (O): $O(\log n)$**

Result

The Recursive Binary Search algorithm was implemented successfully in Python. The program searches a sorted array by repeatedly dividing it into halves and displays the index if the element is found; otherwise, it displays "Element not found". This fulfills the objective of the experiment.